

ROHIT DUVVURU

EMBODIED AI ROBOTICS ENGINEER

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EDUCATION

Bachelor of Science in Computer Science

San Jose State University, 2023

Master of Science in Computer Science

University of Massachusetts Amherst, 2026

EXPERIENCE

Robotics Research Assistant - UMass Amherst HCR Lab

2026 - Present

- Developed an adaptive **multi-robot autonomy** framework in **ROS 2 Humble** for dynamic formation control, coordinated **navigation**, reconfiguration, autonomous joining/leaving, and teleoperation-aware recovery in **simulation**, with integration targeting **real-world AgileX Limo robots**.
- Built a Gymnasium-compatible Pygame simulation and trained a **GNN-based PPO** policy with per-robot control, agent masking, teleoperation disturbances, collision and goal rewards, and formation-error metrics; prepared deployment to AgileX Limo robots using **mocap localization** and **omnidirectional drive control**.

Embedded Software Intern - iVativ

2025

- Integrated sensors with **Nordic Semiconductor BLE-enabled embedded devices** using **I2C, SPI, and UART** interfaces, enabling wireless transmission of sensor data in an **Edge AI** monitoring system.
- Collected, labeled, and analyzed sensor datasets for Edge AI applications; evaluated and trained **ML** models using **Edge Impulse APIs** for deployment on embedded devices.
- Implemented over-the-air (**OTA**) firmware updates to remotely deploy software and inference model updates to field devices.

TECHNICAL SKILLS

- Robotics/Simulation:** ROS/ROS2, Gazebo, RViz, Pygame, SLAM, IsaacSim, IsaacLab
- ML/Robot Learning:** PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, PPO, Reinforcement Learning (RL), Graph Neural Networks (GNNs)
- Programming:** Python, C/C++, Java, Linux, CMake, JavaScript, SQL
- Autonomy:** Localization, LiDAR, Multi-Robot Systems, State Estimation
- Systems/Hardware:** BLE, UART, I2C, SPI, Sensor Integration, Distributed Systems, Multiprocessing, IPC, AWS
- Relevant Coursework:** Robotics, RL, Artificial Intelligence, Machine Learning, Distributed Systems, Parallel Processing, Data Structures & Algorithms

PROJECTS

Particle Filter Localization for Indoor Robot Pose Estimation

Python | ROS Noetic | Gazebo | RViz | NumPy | SLAM | 2D LiDAR

- Implemented **particle filter robot localization** using 1,000 particles with a **likelihood field** sensor model, odometry-based motion prediction, **LiDAR** measurement updates, and systematic resampling.
- Precomputed a **distance-transform** likelihood field from an **occupancy-grid map** and added 10% random **particle reinjection** to improve computational efficiency and preserve hypothesis diversity.
- Achieved stable pose convergence in approximately 20–40 sensor updates and visualized the **map, particle cloud, laser scan, TF frames**, and estimated pose in **RViz**.

Reinforcement Learning for Autonomous Wall Following

Python | ROS Noetic | Gazebo | RViz | Q-Learning | SARSA | 2D LiDAR

- Formulated right-side **wall following** as a **discrete MDP** using three **LiDAR** sectors, discretized distance states, and forward/left/right motion primitives.
- Implemented and compared **Q-Learning** and **SARSA** over 300 episodes using **epsilon-greedy exploration**, state-based **reward shaping**, and **expert-initialized Q-tables**.
- Demonstrated **navigation** along straight walls, turns, and an unseen isolated wall segment; SARSA achieved smoother convergence and a larger peak accumulated reward than Q-learning.